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Correlations of horizontal and vertical spectral accelerations for the Chilean subduction zone

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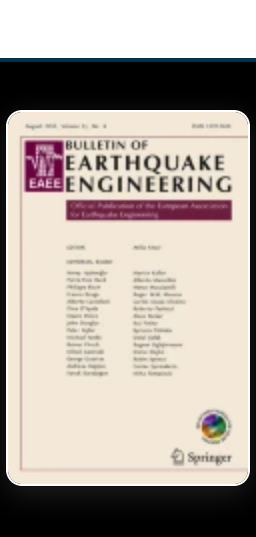

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Abstract

This study develops parametric models for the correlation between vertical and horizontal components of spectral accelerations in the Chilean subduction zone. The models are based on seismic records from national databases and recent ground motion models developed for the region. The effects of magnitude, distance, site conditions, and model selection on correlations are evaluated and found to be negligible, supporting the use of a simplified formulation. The proposed models are compared to existing models developed for other regions, highlighting observed differences in the correlation structure. Despite these differences, the impact on practical applications is limited due to the low sensitivity of hazard estimates to variations in correlation at low correlation values. The results are suitable for use in seismic hazard analyses that incorporate vertical ground motion, enabling more comprehensive assessments for structures sensitive to this component. The study highlights the importance of developing region-specific models and identifies future needs, including expanding local ground motion models to additional intensity measures and refining correlation functions to better capture differences between interface and inslab events.

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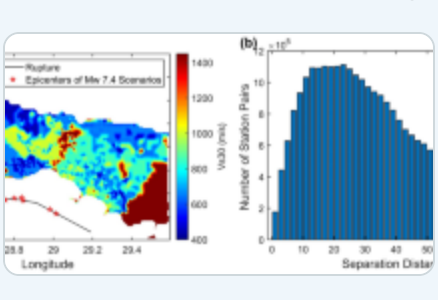

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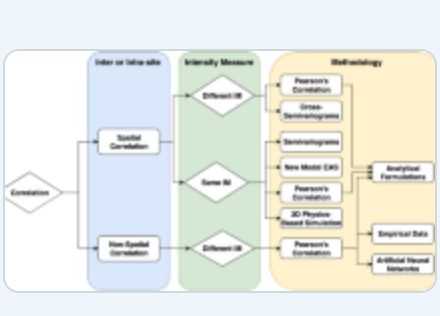

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1 Introduction

During an earthquake, structures are subjected to ground motions in both horizontal and vertical directions. While extensive research and modeling efforts have focused on estimating structural response due to the horizontal components, the vertical component has received considerably less attention. This imbalance is primarily due to the long-standing belief that the vertical component, characterized by lower energy content, has a limited influence on structural damage (Papazoglou and Elnashai 1996). As a result, seismic design codes and engineering practices have historically prioritized horizontal demands, often oversimplifying or neglecting the effects of vertical ground motions. However, research has shown that the vertical component tends to concentrate energy within a narrow frequency band, which can amplify structural response in systems with short vertical natural periods (Papazoglou and Elnashai 1996). These characteristics challenge the assumption of its negligible effect and highlight the need for further investigation, particularly for structures that are sensitive to vertical excitation.

Several studies have examined the influence of vertical ground motion on the structural performance of various systems. Kunnath et al. (2008) investigated the effects of vertical acceleration on highway bridges and overpasses, finding that, under certain conditions, vertical motion can significantly increase shear demands beyond the capacity of columns. Similarly, Bovo and Savoia (2019) studied the behavior of precast concrete structures and concluded that current design approaches underestimate vertical effects, resulting in inadequate safety factors. Similar concerns have been raised for steel structures (Valdés-Vázquez et al. 2021), liquid storage tanks (Kohrangi et al. 2023), and reinforced concrete frames (Kim et al. 2018). These studies consistently indicate that the influence of the vertical component is underestimated in existing structural design standards.

A reliable and consistent seismic hazard analysis that includes the vertical component requires two essential elements: (1) ground motion models (GMMs) that represent vertical motion, and (2) correlation models that quantify the dependence between vertical and horizontal components. GMMs are used to estimate the median and standard deviation of ground motion intensity measures (IMs) at a given site based on parameters such as earthquake magnitude, source-to-site distance, fault mechanism, and local site conditions. These models must be developed specifically for the region of interest, for the relevant intensity measure, component, and seismic source type.

Correlation models, in turn, are used to quantify the statistical dependence between IMs in different directions or at different periods. In the context of vertical motion, correlations between vertical and horizontal (VH), as well as vertical and vertical (VV) spectral accelerations, are of particular interest. These models are critical for a variety of seismic applications, including advanced record selection methods and seismic hazard analyses. For example, the Conditional Mean Spectrum (CMS) method (Baker 2011) relies on such correlations to estimate the expected vertical spectral acceleration conditioned on a target horizontal value. Likewise, vector-valued probabilistic seismic hazard analyses (VPSHA) use correlation models to evaluate the joint probability of exceedance of multiple IMs, such as accelerations in both horizontal and vertical directions or in multiple vibration modes (Bazzurro and Cornell 2002).

Despite the importance of these models, most correlation studies have been developed for crustal earthquakes in active shallow crust tectonic regions (e.g., Baker (2006) and Kohrangi et al. (2020)). These models may not be directly applicable to subduction zones like that of Chile, where both interface and inslab earthquakes contribute significantly to the seismic hazard. The characteristics of vertical ground motions in such environments can differ considerably, underscoring the need for correlation models that are specifically calibrated for local conditions.

In response to this need, the present study develops empirical correlation models for spectral accelerations derived from subduction earthquakes in Chile. The analysis is based on seismic recordings from RENADIC and the Centro Sismológico Nacional (CSN), in conjunction with recently developed Chilean GMMs (Idini et al. 2017; Montalva et al. 2017; Domínguez et al. 2026). The study evaluates the influence of GMM selection, earthquake magnitude, source-to-site distance, and average shear-wave velocity in the upper 30 meters (V_{S30}) on correlation behavior. The proposed models are then compared with existing correlation formulations that incorporate the vertical component to assess regional differences and potential limitations in their applicability to Chilean conditions.

2 Ground motion database

The selection of ground motion records used in this study follows criteria consistent with the applicability ranges of the GMMs employed. Specifically, the GMMs MBR (Montalva et al. 2017), IRRP (Idini et al. 2017), and DHML (Domínguez et al. 2026) define the validity domains within which the predicted intensity measures (IMs) are statistically reliable. These domains guide the selection of records to ensure consistency with the underlying assumptions of each GMM. Table 1 summarizes the validity ranges for the MBR and IRRP models, detailing the limits for key parameters, including earthquake magnitude, source-to-site distance, V_{S30}, and hypocentral depth.

Table 1 Validity ranges of horizontal ground motion models (GMMs). IF: Interface; IS: Inslab

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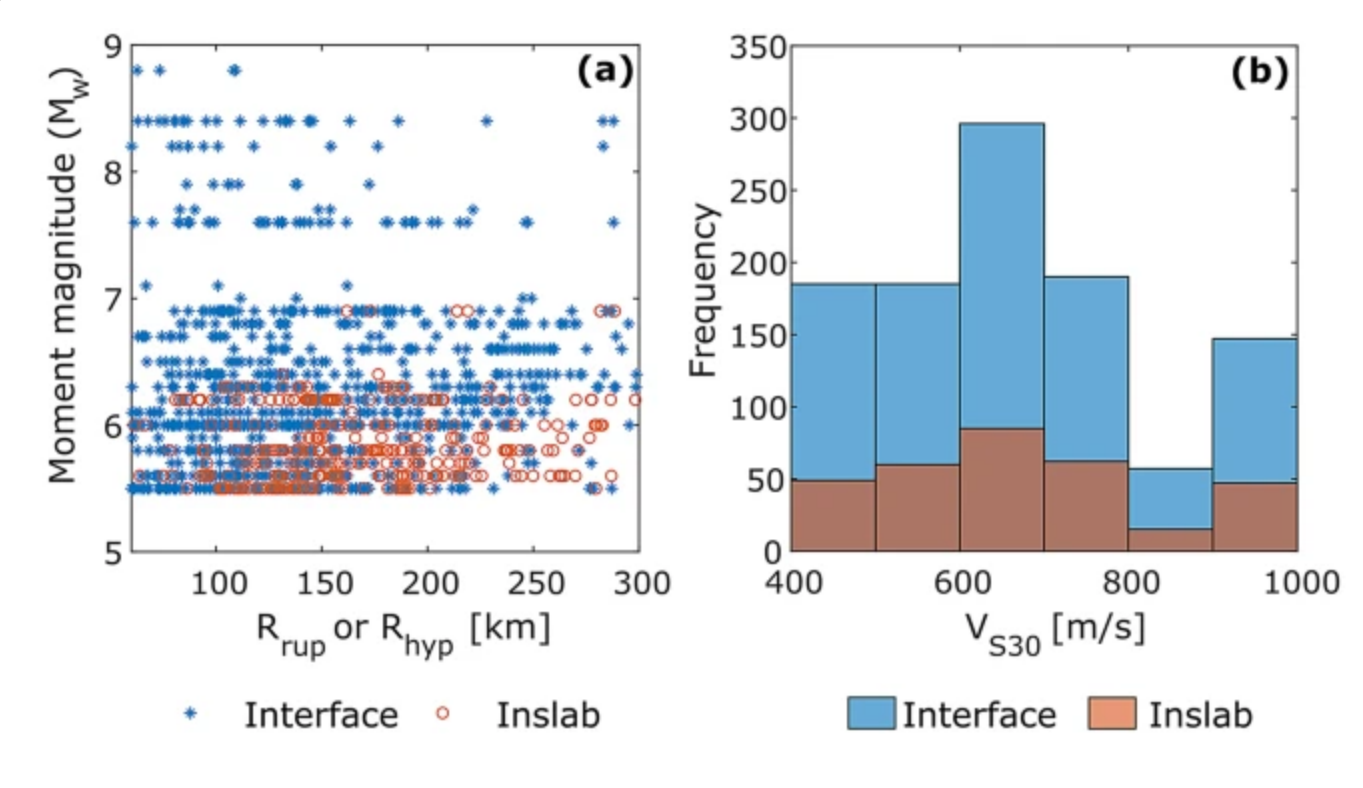
The DHML model is applicable to earthquakes with moment magnitudes greater than 4.0, source-to-site distances between 30 and 150 km for events with magnitudes below 6.0, and up to 300 km for larger events. For interface events, rupture distance (R_{rup}) is used, while hypocentral distance (R_{hyp}) is used for inslab events. Additionally, the model is defined for sites with V_{S30} above 200 m/s, although it shows reasonable performance over a broader range.

Based on these constraints, records were selected from events with magnitudes greater than 5.5, distances between 60 and 300 km, V_{S30} values between 400 and 1000 m/s, and hypocentral depths between 0 and 150 km. This filtering process yielded 1060 interface ground motion records from 144 events, and 318 inslab records from 45 events. These records constitute a subset of the Chilean ground motion database recently published by Bastías et al. (2026). The distribution of these records is shown in Fig. 1. The V_{S30} values are taken from the SIBERRISK database (Castro et al. 2022), which consists of values reported by the CSN, data obtained from the horizontal-to-vertical spectral ratios of records during subduction earthquakes (Yang et al. 2019), and site observations. The database contains a total of 300 stations from RENADIC and CSN. Station specific uncertainty estimates for V_{S30} are not available for all stations; therefore, V_{S30} based subset analyses in subsequent sections are interpreted as indicative of broad site condition trends rather than as a precise quantification of site effects.

Fig. 1



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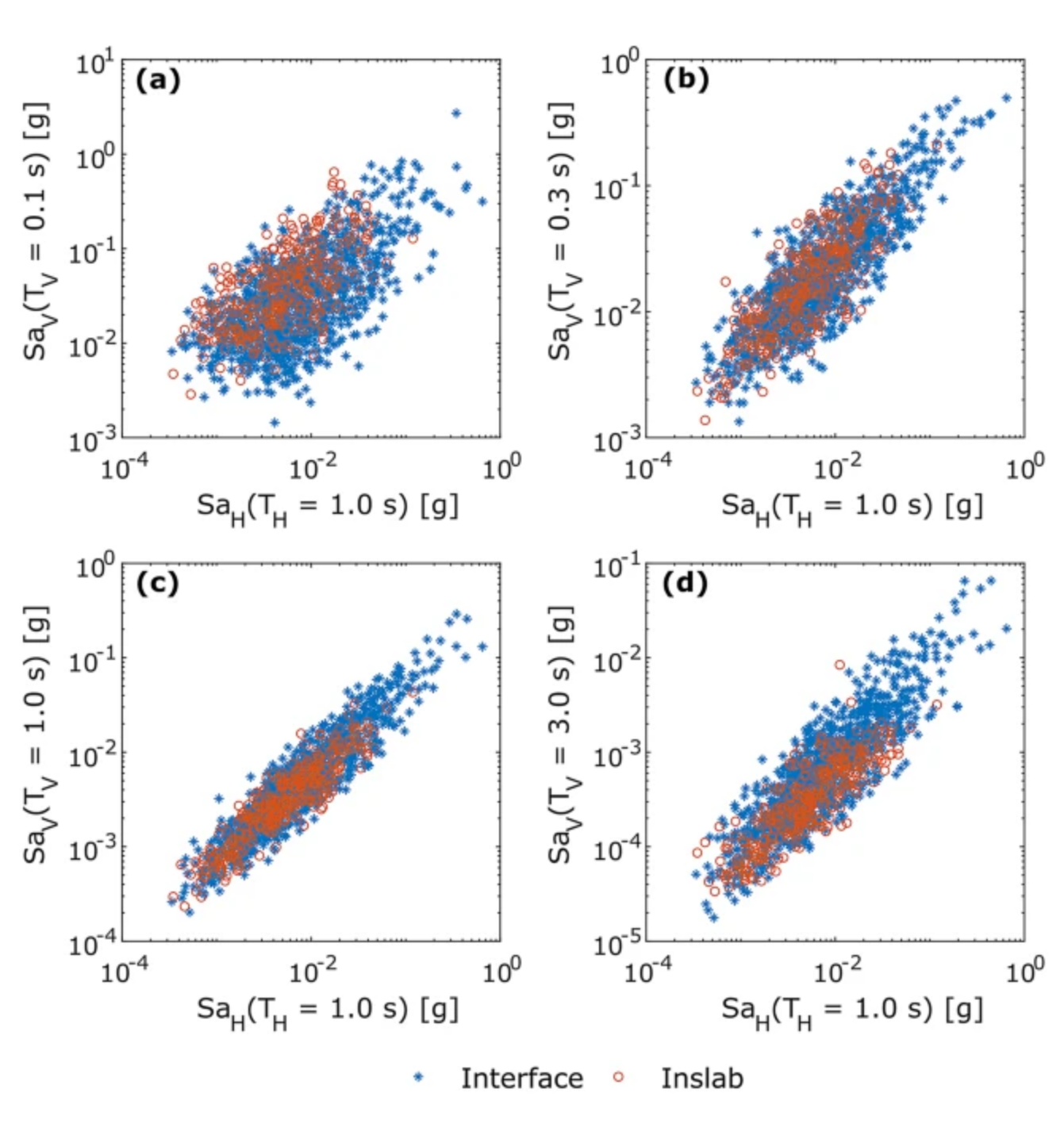
Distribution of selected records for this study as a function of (**a**) magnitude and rupture distance for interface earthquakes, and hypocentral distance for inslab earthquakes; and (**b**) V_{S30}

Vertical spectral accelerations, Sa_V, at periods $T_V = 0.1$, 0.3, 1.0, and 3.0 s present a positive correlation with horizontal spectral acceleration, Sa_H, at a period $T_H = 1.0$ s, as illustrated in Fig. 2. The periods 0.1, 0.3, 1.0, and 3.0 s were selected as representative anchor periods spanning the main spectral range of engineering interest. They capture short-period, intermediate-period, and longer-period responses, including high-frequency ground motion relevant to stiff and low-to-mid-rise structures, the intermediate range commonly used in conditional spectrum applications, and the longer period range relevant to more flexible structures. In all cases, the magnitude of the vertical component increases with increasing horizontal spectral acceleration. The highest correlation values are observed when the periods of the vertical and horizontal components are the same, while the correlation systematically decreases as the difference between the periods increases. This trend is consistent across the range of periods analyzed. The distribution of the data indicates that horizontal spectral acceleration at 1.0 s is statistically related to the amplitude of the vertical component, providing information on its variability across multiple spectral periods without requiring period matching.

Fig. 2



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Vertical versus horizontal component of the selected ground motion records, for a fixed horizontal period of 1.0 s. Four combinations are shown according to the vertical period: (**a**) 0.1 s, (**b**) 0.3 s, (**c**) 1.0 s, and (**d**) 3.0 s

3 Residual variability structure in spectral acceleration ground motion models

The residual terms analyzed in this study arise from the ground motion prediction model defined in Eq. (1) for a given intensity measure, IM:

$$\ln IM = \mu_{\ln IM}\left(R, M, \theta\right) + \delta B + \delta W \tag{1}$$

where $\mu_{\ln IM}\left(R, M, \theta\right)$ is the mean of the natural logarithm of IM, which is a deterministic function of the source-to-site distance, R, event magnitude, M, and other parameters (e.g., site conditions), θ. The terms δB, and δW are the between- and within-event residuals, which follow a zero-mean normal distribution. The standard deviation of δB is τ, whereas the standard deviation of δW is ϕ. This model assumes that δB and δW are independent.

The between-event residual term δB represents a common variability to all recordings from the same earthquake and captures differences between seismic events. The within-event residual term, δW, varies for each record within a given event and accounts for site-to-site variability.

Modern GMMs typically provide direct estimates of the between- and within-event standard deviations, τ and ϕ. In this study, δB and δW residuals are estimated using the maximum likelihood approach (Abrahamson and Youngs 1992). The between-event residual is estimated for each event r by averaging the differences between the observed logarithmic intensity measures and the predicted mean values across all recording stations (Eq. (2)).

$$\widehat{\delta B}_r = \frac{\sum_{s=1}^{n_r} \ln IM_{sr} - \mu_{\ln IM}\left(R_{sr}, M_r, \theta_{sr}\right)}{\frac{\phi^2}{\tau^2} + n_r} \tag{2}$$

where IM_{sr} denotes the observed intensity measure at station s during event r, n_r is the number of recordings associated with that event, and $\mu_{\ln IM}$ is the predicted logarithmic mean at station s during event r, computed from the GMM.

Once the between-event residual is obtained, the within-event residual for each recording can be estimated by subtracting both the predicted mean and the between-event residual from the observed value, as shown in Eq. (3):

$$\widehat{\delta W}_{sr} = \ln IM_{sr} - \mu_{\ln IM}\left(R_{sr}, M_r, \theta_{sr}\right) - \widehat{\delta B}_r \tag{3}$$

The median prediction coefficients of the original GMMs were kept fixed. However, to avoid potential biases in correlation estimates due to the use of a filtered subset, the residuals were post-processed through a detrending procedure. This involved removing mean trends of δB with respect to M, and δW with respect to $R$and V_{S30} (see Supplementary Material for residual diagnostics before and after the detrending process). Additionally, empirical values of τ and ϕ were re-computed from the detrended residuals.

Several GMMs have been developed for the Chilean subduction zone to predict horizontal ground motion intensity measures (Idini et al. 2017; Montalva et al. 2017). For the vertical component Domínguez et al. (2026) developed a GMM for spectral accelerations based on the functional form proposed in BCH16 (Abrahamson et al. 2016) and calibrated using Chilean subduction events.

To date, the only GMM available for the vertical component of spectral acceleration specifically for the Chilean subduction context is the one currently published for this purpose (Domínguez et al. 2026). Since correlation models require consistent input from GMMs, and due to the absence of other vertical-component models, the present study focuses exclusively on developing correlation models for spectral accelerations.

4 Correlation models

To estimate the correlation coefficients between ground motion IMs, this study adopts the Pearson correlation coefficient, defined in Eq. (4) (Neter et al. 1996):

$$\rho_{A,B} = \frac{\sum_{i=1}^{n}\left(A_i - \bar{A}\right)\left(B_i - \bar{B}\right)}{\sqrt{\sum_{i=1}^{n}\left(A_i - \bar{A}\right)^2 \sum_{i=1}^{n}\left(B_i - \bar{B}\right)^2}} \tag{4}$$

Here, $\rho_{A,B}$ represents the correlation between the random variables A and B.

In this study, normalized residuals ε_H and ε_V (total normalized residuals), $\widetilde{\delta B}_H$ and $\widetilde{\delta B}_V$ (between-event normalized residuals), and $\widetilde{\delta W}_H$ and $\widetilde{\delta W}_V$ (within-event normalized residuals) are used as A and B in Eq. (4). The subscript H or V represents horizontal and vertical orientations, respectively. Equation (5) defines the normalized residuals:

$$\varepsilon_{sr} = \frac{\widehat{\delta B}_r + \widehat{\delta W}_{sr}}{\sqrt{\tau^2 + \phi^2}} \tag{5a}$$

$$\widetilde{\delta B}_r = \frac{\widehat{\delta B}_r}{\tau} \tag{5b}$$

$$\widetilde{\delta W}_{sr} = \frac{\widehat{\delta W}_{sr}}{\phi} \tag{5c}$$

Since in ground motion prediction models the only source of uncertainty in the residual decomposition is ε, the correlation between two IMs can be approximated by the correlation between their normalized residuals (Bradley 2012).

For the horizontal component definition, Sa_H is taken as the geometric mean of the 5%-damped spectral accelerations of the two as-recorded horizontal orientations. This definition is consistent with the horizontal component definition adopted in the MBR and IRRP models. Orientation-independent measures, such as RotD50 or RotI50, and single-component horizontal ordinates are not used. However, previous studies (e.g., Baker and Jayaram 2008) have shown that the dependence structure of residuals is relatively insensitive to the specific definition of the horizontal component.

4.1 Parametric fit to correlation models

The parametric representation of correlation coefficients between spectral acceleration components is carried out using functional forms previously proposed in the literature. For vertical–vertical correlations, $Sa_V - Sa_V\ (VV)$, the functional form is adopted from a previous study specifically developed for this type of correlation (Baker and Jayaram 2008). This model expresses the correlation between vertical spectral accelerations at two different periods, $\rho_{VV}\left(T_1, T_2\right)$, as a piecewise function based on the minimum and maximum of the periods T_1 and T_2. It includes three conditional expressions—denoted as C_1, C_2, and C_3—whose full formulation is provided in Eqs. (6), (7), (8) and (9). The parameters a, b, c, d, e, f of this function are empirically calibrated using observational data.

$$\rho_{VV}\left(T_1, T_2\right) = \begin{cases} C_2, & T_{max} < b \\ C_1, & T_{max} \geq b and T_{min} > b \\ \min\left(C_2, C_3\right), & b \leq T_{max} < f and T_{min} \leq b \\ C_3, & \text{otherwise} \end{cases} \tag{6}$$

$$C_1 = 1 - \cos\left(\frac{\pi}{2} - a \cdot \ln\left(\frac{T_{max}}{\max\left(T_{min}, b\right)}\right)\right) \tag{7}$$

$$C_2 = \begin{cases} 1 - c \cdot \left(1 - \frac{1}{1+\exp\left(100 \cdot T_{max} - d\right)}\right)\left(\frac{T_{max} - T_{min}}{T_{max} - e}\right), & T_{max} < f \\ 0, & T_{max} \geq f \end{cases} \tag{8}$$

$$C_3 = C_1 + 0.5\left(\sqrt{C_1} - C_1\right)\left(1 + \cos\left(\frac{\pi \cdot T_{min}}{b}\right)\right) \tag{9}$$

For vertical–horizontal correlations, $Sa_V - Sa_H\ (VH)$, one of the models employed in this study is based on a functional form previously developed for this type of correlation (Kohrangi et al. 2020). When the periods are equal ($T_V = T_H = T$), the correlation $\rho_{VH,0}\left(T\right)$ is defined through a continuous function dependent on a threshold parameter f_1, as shown in Eq. (10). In the general case where $T_V \neq T_H$, a scaling function is introduced based on the maximum and minimum periods, along with an auxiliary function C, detailed in Eqs. (11) and (12), respectively.

$$\rho_{VH,0}\left(T\right) = \begin{cases} a_1 + b_1 \cdot \cos\left(\frac{\pi}{2} + \pi \cdot \frac{\ln\left(\frac{\max\left(T, c_1\right)}{d_1}\right)}{e_1}\right), T \leq f_1 \\ g_1 + h_1 \cdot \cos\left(-i_1 \cdot \pi \cdot \ln\left(\frac{T}{f_1}\right)\right), T > f_1 \end{cases} \tag{10}$$

$$\rho_{VH}\left(T_V, T_H\right) = \rho_{VH,0}\left(T_{\max}\right) \cdot \begin{cases} 1 - a_2 \cdot C + a_2 \cdot C \cdot \cos\left(\pi \cdot \frac{In\left(\frac{T_{\min}}{T_{\max}}\right)}{In\left(\frac{T_{\max}}{b_2}\right)}\right), T_{\min} > b_2 \\ 1 - c_2 \cdot C - d_2 \cdot C \cdot \cos\left(\pi \cdot \frac{In\left(\max\left(\frac{T_{\min}}{b_2}, e_2\right)\right)}{In\left(\max\left(\frac{T_{\max}}{b_2}, f_2\right)\right)}\right), |T_{\min} \leq b_2 \end{cases} \tag{11}$$

$$C = 1 - \cos\left(\left(\frac{\pi}{g_2}\right) \cdot \min\left(h_2, \ln\left(\max\left(\frac{T_{max}}{b_2}, i_2\right)\right)\right)\right) \tag{12}$$

Since empirical correlation values exhibit heteroscedasticity—meaning their standard error depends on the magnitude of ρ—the Fisher z-transformation is applied to stabilize the variance before curve fitting, as defined in Eq. (13). This transformation allows for a robust non-linear least squares adjustment in the transformed space, by minimizing the squared differences between empirical and modeled z-values. As a result, the calibration of model parameters yields statistically consistent and reliable correlation predictions across a wide range of periods.

$$z = \frac{1}{2} \cdot \ln\left(\frac{1+\rho}{1-\rho}\right) \tag{13}$$

The model parameters (Baker and Jayaram 2008; Kohrangi et al. 2020) are estimated using a weighted nonlinear least squares method applied to the Fisher-transformed empirical correlations $\rho_{i,j}$, obtained from Eq. (4). The modeled correlations $\hat{\rho}_{i,j}$ are evaluated using Eqs. (6), (10), or (11). The fitting procedure minimizes the squared differences between transformed empirical and theoretical values across all period pairs $\left(T_i, T_j\right)$, allowing consistent calibration of the model parameters, as follows:

$$\min \sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}\left(\frac{1}{2} \cdot \ln\left(\frac{1+\rho_{i,j}}{1-\rho_{i,j}}\right) - \frac{1}{2} \cdot \ln\left(\frac{1+\hat{\rho}_{i,j}}{1-\hat{\rho}_{i,j}}\right)\right)^2 \tag{14}$$

For each period pair $\left(T_i, T_j\right)$, the variance of the Fisher transformed correlation was taken as $1/\left(n_{ij} - 3\right)$, where $n_{ij}$is the number of usable observations available for that period pair. Accordingly, the fit was weighted using $w_{ij} = n_{ij} - 3$, giving lower influence to period pairs with fewer observations and reducing the effect of less stable correlation estimates, particularly for data limited cases such as inslab events.

5 Results

5.1 Correlation calculations

The empirical correlation results computed using Eq. (4) are shown in Fig. 3, which displays the correlation patterns for different pairs of GMMs for both interface and inslab earthquake scenarios. In general, correlations between vertical-vertical components are consistently higher than those between vertical-horizontal components. Moreover, correlation values tend to decrease as the difference between the vibration periods increases. While the VV correlations are symmetric with respect to the period axes, the VH correlations exhibit noticeable asymmetry. Two possible hypotheses may explain this asymmetry: it could be related to physical effects associated with wave propagation, or it could reflect differences in the regression procedures adopted by the horizontal and vertical GMMs. Targeted tests for these hypotheses are beyond the scope of this study. Nevertheless, the largest discrepancies are observed in period combinations where correlation values are inherently low. As a result, the adoption of a symmetric or asymmetric model for VH correlations is unlikely to produce significant differences in most practical applications, particularly in engineering analyses focused on strongly correlated period ranges.

Fig. 3

Full size image ›

Residual empirical correlations $\rho_{VH}\ (T_V, T_H)$ for subduction interface earthquakes using the GMM pairs MBR–DHML (**a**) and IRRP–DHML (**b**). Panels (**c**) and (**d**) show $\rho_{VH}\ (T_V, T_H)$ for inslab events with the same GMM pairs. Panels (**e**) and (**f**) present vertical-vertical residual correlations $\rho_{VV}\ (T_{V1}, T_{V2})$ for interface and inslab events, respectively, computed using the DHML model

To support a detailed examination of the correlation structure, Figure 4 presents the computed correlations at two fixed periods (0.1 and 1.0 s), as functions of the varying secondary period. The results are accompanied by their corresponding 95% confidence intervals. Distinct differences are observed between interface and inslab seismic events, particularly at shorter periods. In all three pairs of GMMs evaluated (MBR-DHML, IRRP-DHML, and DHML-DHML), interface events generally exhibit higher correlation values than their inslab counterparts. This systematic behavior is indicative of underlying differences in seismotectonic characteristics, such as rupture process, stress drop, and radiation pattern, associated with each type of subduction earthquake. The observed divergence in correlation levels highlights the need to employ distinct correlation models for interface and inslab sources. Adopting separate models improves the physical representativeness of ground motion simulations and reduces epistemic uncertainty in seismic hazard assessments. These findings emphasize the importance of disaggregated modeling strategies in future research and engineering applications.

Fig. 4

Full size image ›

Correlation coefficients $\rho_{VH}\ (T_V, T_H)$ and $\rho_{VV}\ (T_{V1}, T_{V2})$ for subduction earthquakes in Chile. Panels (**a**)–(**b**) show ρ_{VH} using the MBR and DHML ground motion models (GMMs), and (**c**) using the IRRP and DHML GMMs, for periods T_H = 0.1 s and 1.0 s in panels (**a**) and (**b**). Also, in panel (**c**) for periods T_{V1} = 0.1 s and 1.0 s. Each panel includes results for interface and inslab events, with shaded bands indicating 95% confidence intervals

5.2 Comparison with previous models

Empirical correlation results are compared in Fig. 5 with analytical models previously developed for crustal earthquakes (Baker 2006; Kohrangi et al. 2020). A systematic comparison reveals that both VH and VV empirical correlations obtained for subduction interface and inslab events tend to be higher than those reported for crustal earthquakes over a broad range of spectral periods. This behavior is consistent with findings from previous studies conducted in subduction settings (Jayaram et al. 2011; Candia et al. 2020). It has been suggested that the discrepancies observed between subduction and crustal earthquake correlations may be attributed to systematic differences in the average frequency content of ground motions, since this parameter influences both spectral acceleration values and their intercomponent correlations (Jayaram et al. 2011). Moreover, the importance of accurately accounting for site-specific conditions has been emphasized, as local site effects can significantly modify the observed correlation patterns (Candia et al. 2020). These findings collectively underscore the need to develop ground motion correlation models specifically tailored to subduction environments, rather than relying on models calibrated exclusively using crustal earthquake data.

Fig. 5

Full size image ›

Comparison of empirical VH and VV correlation coefficients, $\rho_{VH}\ (T_V, T_H)$ and $\rho_{VV}\ (T_{V1}, T_{V2})$, obtained in this study for subduction earthquakes, against previously proposed models developed for crustal events. Panels (**a**) and (**b**) show the empirical $Sa_V - Sa_H$ correlations for interface and inslab events, respectively, for spectral periods $T_H = 0.1s$ and $1.0s$, plotted as a function of T_V. Panels (**c**) and (**d**) display the $Sa_V - Sa_V$ correlations for interface and inslab events, respectively, for the same reference periods. In all cases, the empirical results are compared against models proposed in previous studies (Baker 2006; Kohrangi et al. 2020)

5.3 Assessment of the influence of GMM, magnitude, distance, and V_{S30} on correlations

To evaluate the impact of GMM selection on VH correlation estimates, Fig. 6 presents the computed correlations using two different horizontal GMMs applied to identical spectral periods. The results show that the use of different GMMs—specifically, those developed for subduction zones (Idini et al. 2017; Montalva et al. 2017)—produces highly similar VH correlation values across all periods, for both interface and inslab earthquakes. This consistency suggests that the choice of GMM has a limited effect on correlation outcomes when models are properly calibrated to regional datasets. Slight deviations are observed in interface events for periods greater than 1.0 s and in inslab events for periods shorter than 0.3 s. These differences may reflect variations in the functional forms, period-dependent standard deviations, or site amplification terms implemented in each GMM. Overall, the results support the notion that VH correlations are predominantly governed by the variability inherent in recorded ground motions, rather than by differences in median spectral acceleration predictions between GMMs. This observation is consistent with previous findings (Baker and Jayaram 2008; Candia et al. 2020; Kohrangi et al. 2020), which emphasize that empirical correlation structures are largely insensitive to the specific GMM used for horizontal motion estimation.

Fig. 6

Full size image ›

Comparison of VH correlation coefficients obtained using the MBR and IRRP ground motion models for subduction earthquakes. Panels (**a**) and (**c**) correspond to interface events, and panels (**b**) and (**d**) to inslab events. Panels (**a**) and (**b**) show results for $T_H = 0.1s$, while panels (**c**) and (**d**) correspond to $T_H = 1.0s$. In all cases, 95% confidence intervals are indicated

The influence of magnitude, site-to-source distance, and V_{S30} on VH correlations for interface earthquakes is illustrated in Fig. 7. This type of analysis was not extended to inslab events due to an insufficient number of ground motion records available for subset evaluation ($N = 318$), which limits statistical reliability. This underscores the distinction in the levels of support for our results: while the dataset is sufficient to characterize the primary correlation differences between interface and inslab events, it does not yet allow for a robust quantification of secondary parametric dependencies (e.g., M, R, or V_{S30}) for the inslab case. The results for interface events reveal a mild dependency of the correlation structure on all these parameters, consistent with previous observations reported in the literature (Baker and Bradley 2017; Kohrangi et al. 2020). In this study, the apparent sensitivity may be partially attributed to the reduced sample sizes in several subsets, particularly those exceeding a magnitude of 7.5 or corresponding to large site-to-source distances. Some of these subsets contain fewer than 100 records, resulting in increased uncertainty and variability in the estimated correlation coefficients. The effect is especially pronounced in the distance-based grouping, where a greater number of intervals results in a lower number of records per subset. Despite these observations, the magnitude of the variations remains limited and unlikely to introduce significant biases in engineering applications. Therefore, adopting correlation models that do not explicitly account for magnitude, distance, or V_{S30} is justified in most practical contexts, as it simplifies implementation without compromising reliability.

Fig. 7

Full size image ›

Correlation coefficients $\rho_{VV}\ (T_{V1}, T_{V2})$ and $\rho_{VH}\ (T_V, T_H)$ for interface earthquakes, evaluated across different subsets of magnitude, rupture distance (R_{rup}), and near-surface shear-wave velocity (V_{S30}). Panel (**a**) shows the dependence of correlations on magnitude, panel (**b**) on rupture distance, and panel (**c**) on V_{S30}. VV and VH correlations are shown together in each panel for comparison

5.4 Parametric correlation models for $\mathbf{Sa_V - Sa_H}$ and $\mathbf{Sa_V - Sa_V}$

Parametric models were fitted to the empirical correlation data for both interface and inslab subduction earthquakes, employing the functional forms proposed for crustal events in previous studies (Baker and Jayaram 2008). The model for VH correlations with $T_V = T_H$ is expressed in Eq. (10), and the resulting coefficients are shown in Table 2. This model provides a continuous representation of VH correlations across different spectral periods $T_V = T_H = T$. Figure 8 presents the resulting correlation model, plotted alongside the corresponding empirical data for each earthquake type. The comparison demonstrates good agreement between empirical observations and model predictions, supporting the use of these parametric formulations in seismic hazard analyses for subduction zones.

Table 2 Coefficients for the vertical-horizontal correlation model with equal periods ($T_V = T_H$) (Kohrangi et al. 2020). See Eq. (10)

Full size table ›

Fig. 8

Full size image ›

VH correlation model, $\rho_{VH,0}$, developed in this study for subduction earthquakes under the condition $T_H = T_V = T$. The figure presents both empirical correlation values and the corresponding fitted model for interface and inslab events

Figure 9a and 9b show the fitted model for VH correlations with $T_V \neq T_H$. This model is expressed in Eqs. (11) and (12), with coefficients presented in Table 3. Comparing the resulting model against the empirical correlations of Figs. 3a to 3d, it is possible to observe that the model originally proposed for crustal earthquakes (Kohrangi et al. 2020) provides a satisfactory fit to the empirical VH correlations for interface events. However, it fails to accurately reproduce the correlation patterns observed for inslab earthquakes, particularly as the period separation increases. This discrepancy likely stems from differences in frequency content and rupture characteristics between the two earthquake types. Moreover, it should be noted that, compared to interface events, empirical correlations obtained for inslab events have higher uncertainty due to the limited sample size. Despite these limitations, the impact on practical applications is expected to be minimal, as the most significant deviations occur at low correlation levels, where small absolute differences (i.e., on the order of 0.1) have limited effect on the shape of joint probability distributions.

Fig. 9

Full size image ›

Correlation models developed in this study for subduction earthquakes. Panels (**a**) and (**b**) show the $Sa_V - Sa_H$ correlation model, $\rho_{VH}\ (T_V, T_H)$, for interface events and inslab events, respectively. Panels (**c**) and (**d**) present the $Sa_V - Sa_V$ correlation model, $\rho_{VV}\ (T_{V1}, T_{V2})$, for interface and inslab events, respectively

Table 3 Coefficients for the vertical-horizontal correlation model with different periods ($T_V \neq T_H$) (Kohrangi et al. 2020). See Eqs. (11) and (12)

Full size table ›

VV correlation data shown in Figs. 3e and 3f were also fitted using the model expressed in Eqs. (6) to (9) (Baker and Jayaram 2008), obtaining the coefficients presented in Table 4. Figures 9c and 9d illustrate the correlation models developed for interface and inslab earthquakes, respectively. Consistent with the observations for VH correlations, the functional form originally calibrated using crustal earthquake datasets provides a satisfactory approximation of empirical correlations for interface subduction earthquakes. However, for inslab events, the model exhibits some discrepancies, particularly when the difference between periods increases. These deviations are likely attributable to differences in source characteristics and frequency content. Nevertheless, as previously discussed, such discrepancies remain within a range that is unlikely to affect practical applications, given that they occur at low correlation levels.

Table 4 Coefficients for the vertical-vertical correlation model ($Sa_V - Sa_V$) (Baker and Jayaram 2008). See Eqs. (6) to (9)

Full size table ›

6 Example of application

To illustrate the practical implications of the proposed correlation models, conditional mean vertical spectra were computed for two representative subduction scenarios. In both cases, the vertical spectrum was conditioned on the horizontal spectral acceleration at $T_H = 1.0$s with $\varepsilon_H = 1$, so that differences among the resulting conditional spectra arise only from the adopted correlation models. Figure 10 compares the unconditional vertical spectrum with conditional spectra computed using the model of Kohrangi et al. (2020), the empirical correlation coefficients directly, and the proposed model.

Fig. 10

Full size image ›

Comparison of unconditional and conditional mean vertical response spectra for two representative subduction scenarios: (**a**) interface scenario with M = 8.5, R = 90 km, and V_{S30} = 760 m/s; (**b**) inslab scenario with M = 7.5, R = 90 km, hypocentral depth of 90 km and V_{S30} = 760 m/s. The black curve shows the unconditional vertical spectrum, whereas the red dashed, green dotted, and blue curves show the conditional spectra obtained using the correlation models of Kohrangi et al. (2020), the empirical correlation values, and the proposed correlation model, respectively

The results show that the proposed model closely reproduces the empirical conditional spectra across the entire period range while maintaining a smooth period-dependent behavior. The close agreement between the conditional spectrum derived from the proposed model and those calculated using empirical correlation coefficients validates the proposed parametric formulations. This consistency demonstrates that the minor discrepancies previously noted at low correlation levels, particularly at large period separations, have a negligible impact on practical engineering metrics. Consequently, these results confirm that the proposed models effectively capture the essential features of the vertical-to-horizontal dependence structure in the Chilean subduction zone.

Notably, the model by Kohrangi et al. (2020), which was calibrated using crustal earthquake datasets, consistently yields lower conditional spectral ordinates compared to the subduction-specific empirical data and our proposed model. This underestimation is directly related to the higher VH correlations observed in the Chilean subduction zone compared to active shallow crustal regions (see Fig. 5). While the absolute differences in spectral shape may appear moderate, the use of a region-specific correlation model ensures a more reliable and physically consistent characterization of vertical demands, which is essential for the design of structures sensitive to vertical ground motion effects.

7 Conclusions

In this study, correlation models were developed for the vertical-horizontal and vertical-vertical component spectral accelerations, specifically for subduction earthquakes in Chile. The models were calibrated using a comprehensive dataset of seismic records provided by RENADIC and the Centro Sismológico Nacional (CSN). GMMs employed in the analysis include models specifically developed for the Chilean subduction zone (Idini et al. 2017; Montalva et al. 2017; Domínguez et al. 2026), namely MBR, IRRP, and DHML, respectively. These models ensure that the empirical correlations are representative of regional seismotectonic conditions and local site effects.

When evaluating the influence of GMM selection, earthquake magnitude, source-to-site distance, and V_{S30} on the behavior of correlation models, slight dependencies were identified in the empirical data. However, these dependencies were found to be statistically weak and not significant enough to affect practical engineering applications. As a result, the use of a simplified correlation model—calibrated with a GMM developed for the Chilean subduction zone (Montalva et al. 2017) and excluding magnitude, distance, and V_{S30} as predictive variables of the correlation—is recommended. This approach enhances model applicability while maintaining consistency with observed data.

Some differences were identified when calibrating empirical correlation data against models developed previously (Baker and Cornell, 2006; Kohrangi et al. 2020), particularly in the case of VV and VH correlations for inslab earthquakes within the Chilean subduction zone. These discrepancies are more pronounced at large period separations and may reflect fundamental differences in source characteristics or wave propagation mechanisms. Nevertheless, their impact on practical applications is expected to be limited, as the accuracy tends to decrease at low correlation values, where deviations have a minimal influence on the resulting joint probability distributions. These findings highlight the need for further investigation into the physical causes of the observed differences. Furthermore, given the limited number of inslab recordings currently available in the Chilean subduction zone compared to interface events, the proposed models for this source should be viewed as capturing primary regional trends while carrying a higher degree of epistemic uncertainty.

The results derived from the proposed correlation models offer essential input for advanced seismic hazard assessment methodologies that explicitly incorporate the vertical component of ground motions. These applications include the development of conditional mean spectra (Baker 2011), conditional spectra, and probabilistic vector-valued seismic hazard analysis (Bazzurro and Cornell 2002). Integrating these approaches enables a more robust and realistic characterization of seismic demands, particularly for structural systems that are sensitive to vertical ground motion effects. As such, the presented models contribute to improving the reliability of seismic design and risk mitigation strategies in subduction environments.

The proposed ρ_{VH} model is calibrated using Sa_H defined as the geometric mean of the 5%-damped spectral accelerations of the two as-recorded horizontal components. Therefore, the model is directly applicable to this horizontal intensity measure, consistent with the Chilean GMMs used in this study. The ρ_{VV} correlations are independent of the horizontal component definition. Furthermore, previous studies (e.g., Baker and Jayaram 2008) have shown that the use of alternative horizontal measures, such as RotD50 or RotI50, have a negligible effect on ρ_{VH} values. Nonetheless, users should be aware that the present ρ_{VH} model is specifically calibrated for the geometric mean to maintain consistency with the employed GMMs.

Data availability

The ground motion records used in this study are a subset of the Chilean ground motion database published by Bastías et al. (2026), accessible through DesignSafe at https://doi.org/10.17603/ds2-6gyp-t353. The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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The authors have no relevant financial or non-financial interest to disclose.

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